

4.3 Compositions of Fetch Vectors

Since the fetch vectors represent the accrual of academic capital units by each manuscript, and all the fetch vectors have dimensionality $|M|$, this allows us to collect certain fetch vectors of interest to form certain communities. These collections of fetch vectors would form subspaces within the space of all manuscripts. These collections could be defined by authors, academic fields of research, institutions, research groups, geographic locations, etc. Summary statistics on each collection would give useful information about the interchanges of academic value within each such collection.

The partitioning of the reference graph into distinct communities can allow computations of similar measures on overall communities, since each community can also have citation edges coming from and going to other communities. The in-degrees and out-degrees of such communities will allow measures of collaboration and global impact of specific academic communities. These measures can also be spread out over time to analyze the dynamics of academic interest and progress. In a similar vein, within the communities, connectivity measures would elucidate within community collaborations.

4.4 Community Detection

In order to partition the citation graph into different communities, community detection needs to be performed. However, citation networks are DAGs, where the eigenvalues of the adjacency matrix are all zeros, the eigenvectors belong to the null space of the adjacency matrix and can take complex values. Therefore, in order to detect citation based communities, we make the assumption that the directed relationship is not important for detecting communities within academic works. Following this, we symmetrize the references graph, and then use the spectral analysis detailed in section 3.6.

$$G_W = G_W + G_W^T$$

Since the edges of the references graph are weighted (by default) by the amount of references in each paper, this makes for a natural measure of how strongly the papers are connected to one another, with the strength diminishing as the citing paper cites more papers. Then, the zero, small and large eigenvalues have a similar meaning as in section 3.6

4.5 Centrality Measure

Centrality measures identify whether a node is crucial in connecting different parts of the graph. For references graph, a central node would be the manuscripts that synthesize many past works and add value to conceptual understanding for the newer works. These manuscripts would represent highly influential works in a field, or influential interdisciplinary works across fields. We show how betweenness centrality can be computed here, but other centrality measures, like degree centrality, closeness centrality, or flow-based centrality could also be used in detecting the central nodes. Betweenness centrality for the references graph can be computed as the ratio of shortest directed paths (SP) between 2 manuscripts m_x, m_z which pass through manuscript m_y

$$\text{Centrality}_y^{\text{btw}} = \sum_{y \neq x \neq z} \frac{SP_{x,z}(y)}{SP_{x,z}} \quad (29)$$

4.6 Basic Distribution Metrics

Using the notion of academic capital and relative citation introduced in section 2.1 allows computation of distribution of capital accrual over the space of manuscripts. These metrics can either be computed for individual manuscript M_j cited by manuscripts $\{M_i\}$ to extract the temporal distribution of capital accrual, or extract distributions of capital accrual across different academic communities.

The temporal distribution would quantify test-of-time relevance of the manuscript, while distributions across different academic communities would quantify the breadth of impact of the manuscript (referred to as spatial distribution henceforth, since this would be a span in the space of manuscripts).

Further, distributions could also be computed for different academic communities by using aggregates of fetch vectors. Temporal distributions of capital accrual into academic communities could enable detecting overall interest in the specific academic community, and help formulate birth-death dynamics on communities.

The spatial distribution of capital accrual by a community would inform about the breadth of impact of a community as a whole.

4.7 Powers of References Graph

Compositions of G_W can include the two-step graph (n-step graph, in general), and gram matrix computation. The two-steps adjacency of the weighted graph $\mathbf{G}_W^2(x, y)$ captures the amount of academic capital manuscript m_y contributes to manuscript m_x through 2-steps connections. The row sums would give the impact of manuscript m_x on manuscripts two generations later. Similarly, n -th power of G_W , $\mathbf{G}_W^n(x, y)$ reports the impact of manuscript m_x on m_y n steps/generations later. This allows to evaluate long-range academic impact of manuscripts. When using weighted edges with $w_{x,y} \in [0, 1]$, the influence would diffuse away rapidly, hence persistence of influence over multiple steps of the reference graph would filter out extremely influential works. However, a much more useful operation would be the gram matrices discussed in section 4.8 and section 4.9.

4.8 Gram Matrix of References Graph

The gram-matrix of the references graph shows interesting properties, even in the case when edges are unweighted. The following properties can be observed from the matrix given by, $\mathbf{G}_U^T \mathbf{G}_U$

- when $x = y$, i.e., for diagonal entries, $\mathbf{G}_U^T \mathbf{G}_U(x, x)$ encodes the number of references of the manuscript m_x
- when $x \neq y$, i.e., for non-diagonal entries, $\mathbf{G}_U^T \mathbf{G}_U(x, y)$ gives the number of references manuscripts m_x and m_y have in common. This can be used to estimate a form of *bibliometric coupling*

Further, the gram-matrix of the weighted reference graph $\mathbf{G}_W^T \mathbf{G}_W$ has following related properties:

- when $x = y$, $\mathbf{G}_W^T \mathbf{G}_W(x, x)$ gives the concentration of academic capital outflow, in terms of the citations it contributes, from the manuscript x without considering its influence on other manuscripts
- when $x \neq y$, $\mathbf{G}_W^T \mathbf{G}_W(x, y)$ gives the concentration of academic capital outflow, in terms of the citations it contributes, from the manuscripts m_x and m_y going to some set of common manuscripts M

4.9 Transpose Gram Matrix

The gram-matrix of the transpose of the references graph shows interesting properties too. If the gram matrix of the original graph described concentration of outflow, the gram matrix of the transpose characterizes that of inflow. For the unweighted reference graph, the gram matrix given by, $\mathbf{G}_U \mathbf{G}_U^T$, can be observed to have the following properties:

- when $x = y$, i.e., for diagonal entries, $\mathbf{G}_U \mathbf{G}_U^T(x, x)$ represents the number of citations of the manuscript m_x
- when $x \neq y$, i.e., for off-diagonal entries, $\mathbf{G}_U \mathbf{G}_U^T(x, y)$ represents the number of manuscripts that cite both m_x and m_y

Similarly, for the weighted reference graph, the gram matrix given by, $\mathbf{G}_W \mathbf{G}_W^T$, can be observed to exhibit the following properties:

- when $x = y$, i.e., for diagonal entries, $\mathbf{G}_W \mathbf{G}_W^T(x, x)$ is proportional to the concentration of academic impact influenced by manuscript m_x independently. When expressed as a fraction of the total academic capital accrued by the manuscript, this represents what percentage of that capital is not shared with other manuscripts.
- when $x \neq y$, i.e., for off-diagonal entries, $\mathbf{G}_W \mathbf{G}_W^T(x, y)$ represents the co-concentration of academic impact influenced by manuscripts m_x and m_y . It may be better understood as the fraction of the manuscript m_x 's academic capital that comes from sources which also cite manuscript m_y .

5 Capital Graph

Representation of the graphs as matrices allows for various convenient algebraic operations, not only with the same graph, but also between the reference and shares graph. To simplify the language and operations in this section, we will refer to graphs G , and their adjacency matrix representations $\mathbf{G}$ interchangeably. We will proceed to demonstrate compositions of the graphs above and how such compositions can be interpreted.

5.1 Capital Graph Computation

The Capital graph $G_{\mathbf{x}}$ would have a structure similar to the shares graph G_S , but with the difference that edge weights represent academic capital received from the manuscripts. The capital graph in its condensed form $G'_{\mathbf{x}}$ can be derived from the condensed shares graph ($\mathbf{G}'_S$) and references graph ($\mathbf{G}_W$) adjacency matrices using definitions given in eq. (28) and eq. (10). Formally, we first compute the $\mathbf{x}$ of all the manuscripts $\mathbf{x}_M$ using eq. (28) by right multiplying it with a vector of 1s, $\mathbf{1}_{|M|} = (1, \dots, 1) \in \mathbb{R}^{|M|}$. Then, we element-wise multiply the vector with each column of G'_S , i.e., compute the Hadamard product (denoted $\odot$)

$$\mathbf{x}_M = \mathbf{G}_W \mathbf{1}_{|M|} \quad (30)$$

$$\mathbf{G}'_{\mathbf{x}} = \mathbf{x}_M \odot \mathbf{G}'_S \quad (31)$$

The full square capital graph matrix $\mathbf{G}_{\mathbf{x}}$ can be constructed from its reduced representation $\mathbf{G}'_{\mathbf{x}}$ in a similar way as the full square shares graph $\mathbf{G}_S$ would be constructed. Similar to the shares graph, the cumulative capital owned across all manuscripts can be obtained by a sum along columns of this matrix. (Substitute $C \equiv c$ in eq. (10)).

5.2 Similarities to Shares Graph

The full Capital graph $G_{\mathbf{x}}$ will exhibit the following similarities with the full Shares graph G_S

- **Adjacency Structure :** The adjacency matrix of the capital graph $\mathbf{G}_{\mathbf{x}}$ will have similar structure as that of the shares graph in eq. (12), however each of the edges will represent the capital value contributor c owns on manuscript m , $\mathbf{x}_{m,c}$.
- **Fetch vectors :** Fetch vectors give unnormalized distributions as described in section 3.2, with the difference being here that the distributions would be of $\mathbf{x}$. However, the row and column sums would not add up to 1, since the amount of academic capital accrued by each individual manuscript can vary.
- **Fetch vector compositions :** Similar to the description in section 3.3, composition of fetch vectors on the capital graph represent different portfolios (check section 6.1).
- **Degree Matrix :** The degree matrix has a similar structure as well, but instead of having cumulative shares, the diagonal elements have cumulative capital, for both manuscripts and contributors.
- **Laplacian and Spectral Analysis :** Spectral analysis can be performed on $\mathbf{G}_{\mathbf{x}}$ as well, by normalizing the Laplacian, and using the analysis in section 3.6. The only difference from section 3.6 is that the connectivity structure now is determined in terms of academic capital $\mathbf{x}$.
- **Spanning Trees Ratio** Spanning trees ratio can also be computed based on $\mathbf{x}$ based weightings using eq. (23).

5.3 Two-Step Capital Graph

The capital graph $G_{\mathbf{x}}$ can be further composed with itself to form the two-step capital graph, which has a similar node traversal information as the two-step shares graph, but with the edges weighted by academic capital accrued by the contributors on the manuscripts.

The adjacency matrix is equivalent to the gram-matrix, given by, $\mathbf{G}_{\mathbf{x}}^2 = \mathbf{G}_{\mathbf{x}}^T \mathbf{G}_{\mathbf{x}}$, which will have a block diagonal structure, with two major blocks on the diagonal. While the $\mathbf{M}_{\mathbf{x}}^2$ block will represent the distribution of capital over the manuscripts, similar to the $\mathbf{M}_S^2$ block in section 3.8, the other diagonal block reveals some

interesting quantities.

$$\mathbf{G}_{\mathbf{x}}^2 = \left[\begin{array}{c|c|c|c} \mathbf{M}_{\mathbf{x}}^2 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{x}_A^2 & \mathbf{x}_A \mathbf{x}_P & \mathbf{x}_A \mathbf{x}_R \\ \hline \mathbf{0} & \mathbf{x}_A \mathbf{x}_P & \mathbf{x}_P^2 & \mathbf{x}_P \mathbf{x}_R \\ \hline \mathbf{0} & \mathbf{x}_A \mathbf{x}_R & \mathbf{x}_P \mathbf{x}_R & \mathbf{x}_R^2 \end{array} \right]$$

The $\mathbf{C}_{\mathbf{x}}^2$ block would be composed of 9 blocks again, representing all combinations of starting and ending node categories. We will walk through the first three blocks for traversing from an author node, and remark the values.

- $\mathbf{x}_A^2$ represents traversing to an author node from another author node.

When both nodes are the same, it represents the concentration of capital accrued on manuscripts. If the author has accrued a lot of capital on all of their manuscripts, the value would be high. This would be a measure of lifetime academic productivity of the contributor as an author.

When the source and destination nodes are different, the number represents the academic value of collaboration between the concerned authors. A higher value indicates that manuscripts involving both the contributors as authors has generated large academic capital, and hence higher impact.

- $\mathbf{x}_A \mathbf{x}_P$ represents traversing to a peer reviewer node from an author node.

The diagonal elements would be zero. (Check section 3.8)

The off-diagonal elements represent the academic capital produced the contributor interacted as a peer reviewer with the author. Since we expect the cost of peer reviews to be small compared to author shares, a high value here would usually be indicative of collusion or other anomaly, i.e., a high-risk author. This metric would need to be paired with other metrics, i.e., portfolio risk to make a conclusive decision. In the case that collusion is ascertained, the value would represent the academic cost of collusion (in terms of $\mathbf{x}$).

- $\mathbf{x}_A \mathbf{x}_R$ represents traversing to a replicator node from an author node. The diagonal and off-diagonal elements have similar interpretations as for the $\mathbf{x}_A \mathbf{x}_P$ block.